8.5 Visual Dialog

- a visual dialog (VisDial) dataset containing 10 question-answer pairs on $\sim$120k images, with a total of $\sim$1.2M dialog question-answer pairs, is constructed
- also, a family of neural encoder-decoder models for visual dialog with 3 encoders – Late Fusion, Hierarchical Recurrent Encoder, and Memory Network – and 2 decoders (generative and discriminative) are introduced
- a number of neural visual dialog answerer models are developed
- the model is given as input – an image I, the "ground-truth" dialog history (including the image caption) $H = (\underbrace{C}_{H_0}, \underbrace{(Q_1, A_1)}_{H_1}, \ldots, \underbrace{(Q_{t-1}, A_{t-1})}_{H_{t-1}})$, the question Q_t, and a list of 100 candidate answers $\mathcal{A}_t = \{A_t^{(1)}, \ldots, A_t^{(100)}\}$ – and asked to return a sorting of $\mathcal{A}_t$
- at a high level, all the models follow the encoder-decoder framework, i.e., are factorized into two parts – (1) an encoder that converts the input (I, H, Q_t) into a vector space, and (2) a decoder that converts the embedded vector into an output
- we describe choices for each component next

8.5 Visual Dialog

- **Decoders**: two types of decoders are used:
  - **Generative** (LSTM) decoder: where the encoded vector is set as the initial state of the LSTM RNN language model. During training, the log-likelihood of the ground truth answer sequence is maximized given its corresponding encoded representation (trained end-to-end). To evaluate, the model's log-likelihood scores are used and candidate answers are ranked. Note that this decoder does not need to score options during training. As a result, such models do not exploit the biases in option creation and typically underperform models that do, but it is debatable whether exploiting such biases is really indicative of progress. Moreover, generative decoders are more practical in that they can actually be deployed in realistic applications.
  - **Discriminative** (softmax) decoder: computes dot product similarity between input encoding and an LSTM encoding of each of the answer options. These dot products are fed into a softmax to compute the posterior probability over options. During training, the log-likelihood of the correct option is maximized. During evaluation, options are simply ranked based on their posterior probabilities.

8.5 Visual Dialog

- **Encoders**: three different encoders (listed below) that convert inputs (I, H, Q_t) into a joint representation are developed
- in all cases, I is represented via the ℓ_2-normalized activations from the penultimate layer of VGG-16
- for each encoder E, all possible versions are experimented with: $E(Q_t)$, $E(Q_t, I)$, $E(Q_t, H)$, $E(Q_t, I, H)$ (for some encoders, not all combinations are "valid"; details below)
  - **Late Fusion (LF) Encoder**: In this encoder, we treat H as a long string with the entire history $(H_0, \ldots, H_{t-1})$ concatenated. Q_t and H are separately encoded with two different LSTMs, and individual representations of participating inputs (I, H, Q_t) are concatenated and linearly transformed to a desired size of joint representation.

8.5 Visual Dialog

- **Hierarchical Recurrent Encoder (HRE)**: In this encoder, the intuition that there is a hierarchical nature to the problem is captured – each question Q_t is a sequence of words that need to be embedded, and the dialog as a whole is a sequence of question-answer pairs (Q_t, A_t). Thus, as shown in Figure 11, an HRE model is proposed that contains a dialog-RNN sitting on top of a recurrent block (R_t). The recurrent block R_t embeds the question and image jointly via an LSTM (early fusion), embeds each round of the history H_t, and passes a concatenation of these to the dialog-RNN above it. The dialog-RNN produces both an encoding for this round (E_t in Figure 11) and a dialog context to pass onto the next round. We also add an attention-over-history ("Attention" in Figure 11) mechanism allowing the recurrent block R_t to choose and attend to the round of the history relevant to the current question. This attention mechanism consists of a softmax over previous rounds $(0, 1, \ldots, t-1)$ computed from the history and question+image encoding.

8.5 Visual Dialog

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  - **Memory Network (MN) Encoder**: A MN encoder is developed that maintains each previous question and answer as a "fact" in its memory bank and learns to refer to the stored facts and image to answer the question. Specifically, Q_t is encoded with an LSTM to get a 512-dimensional vector, each previous round of history $(H_0, \ldots, H_{t-1})$ is encoded with another LSTM to get a $t \times 512$ matrix. We compute inner product of question vector with each history vector to get scores over previous rounds, which are fed to a softmax to get attention-over-history probabilities. Convex combination of history vectors using these attention probabilities gives us the "context vector", which is passed through a fully-connected layer and added to the question vector to construct the MN encoding.
- we use a "[encoder]-[input]-[decoder]" convention to refer to model-input combinations
- for example, "LF-QI-D" has a Late Fusion encoder with question+image inputs (no history), and a discriminative decoder
- implementation details about the models are given next

8.5 Visual Dialog

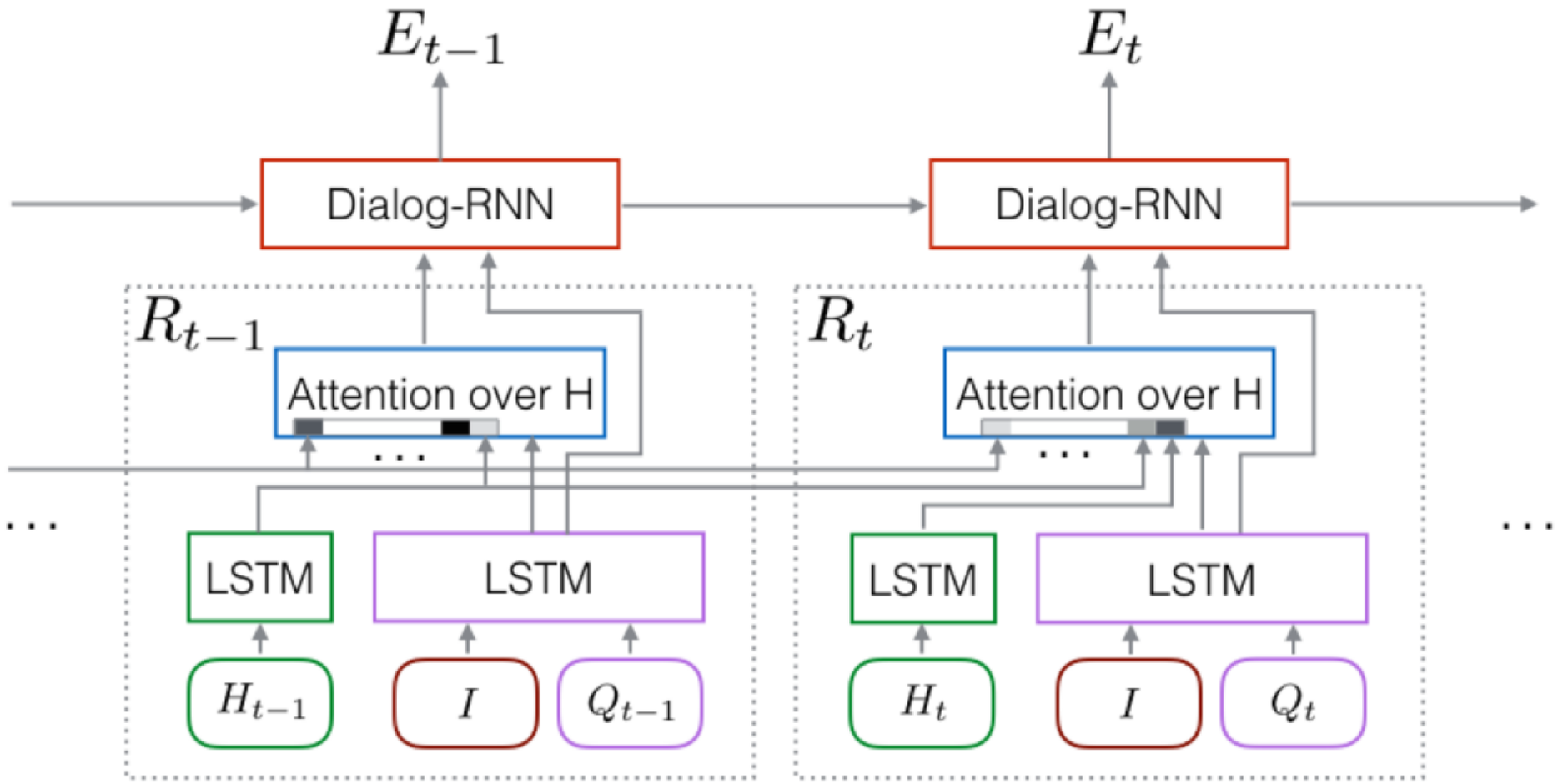


Figure 11: Architecture of Hierarchical Recurrent Encoder (HRE) encoder with attention. At the current round R_t, the model has the capability to choose and attend to relevant history from previous rounds, based on the current question. This attention-over-history feeds into a dialog-RNN along with the question to generate joint representation E_t for the decoder.

8.5 Visual Dialog

- **Late Fusion (LF) Encoder**: the image is encoded with a VGG-16 CNN, the question and concatenated history are encoded with separate LSTMs, and the three representations are concatenated
- this is followed by a fully-connected layer and tanh nonlinearity to a 512-dimensional vector, which is used to decode the response
- Figure 12 shows the model architecture for the LF encoder

8.5 Visual Dialog

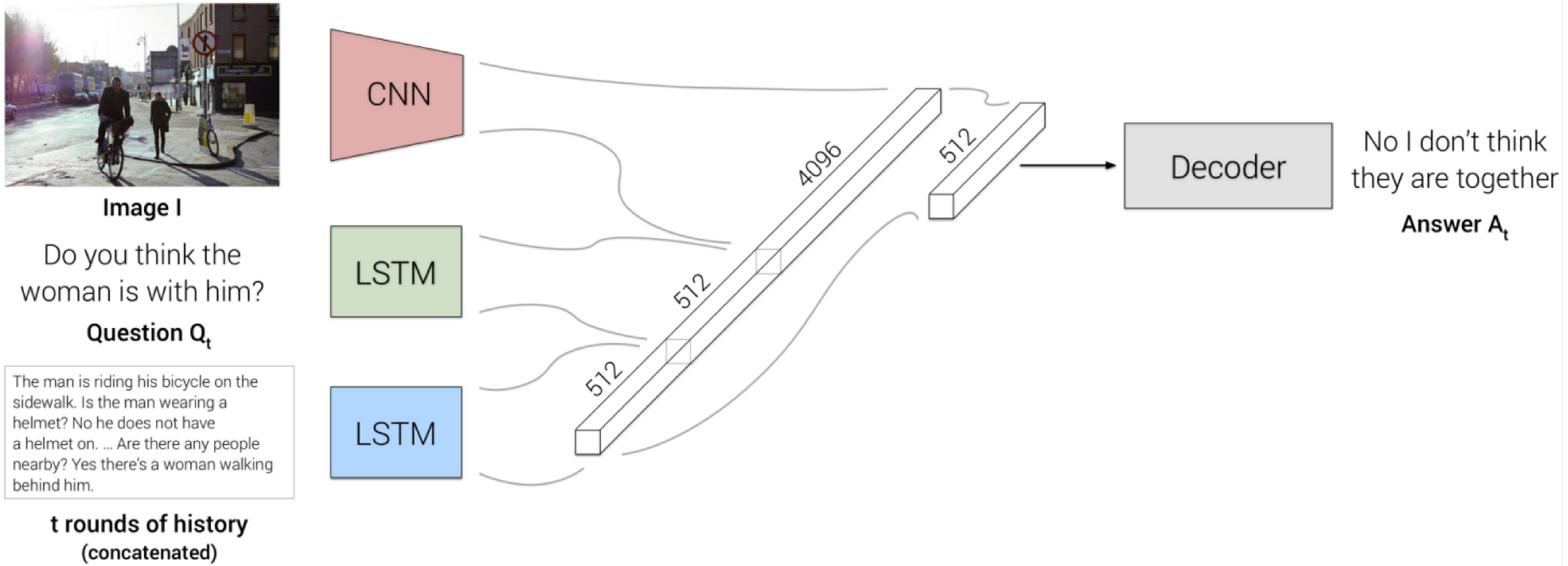


Figure 12: Late Fusion (LF) Encoder.

8.5 Visual Dialog

- **Hierarchical Recurrent Encoder (HRE)**: in this encoder, the image representation from VGG-16 CNN is early fused with the question
- specifically, the image representation is concatenated with every question word as it is fed to an LSTM
- each QA-pair in dialog history is independently encoded by another LSTM with shared weights
- the image-question representation, computed for every round from 1 through t, is concatenated with history representation from the previous round and constitutes a sequence of question-history vectors
- these vectors are fed as input to a dialog-level LSTM, whose output state at t is used to decode the response to Q_t
- Figure 13 shows the model architecture for the HRE

8.5 Visual Dialog

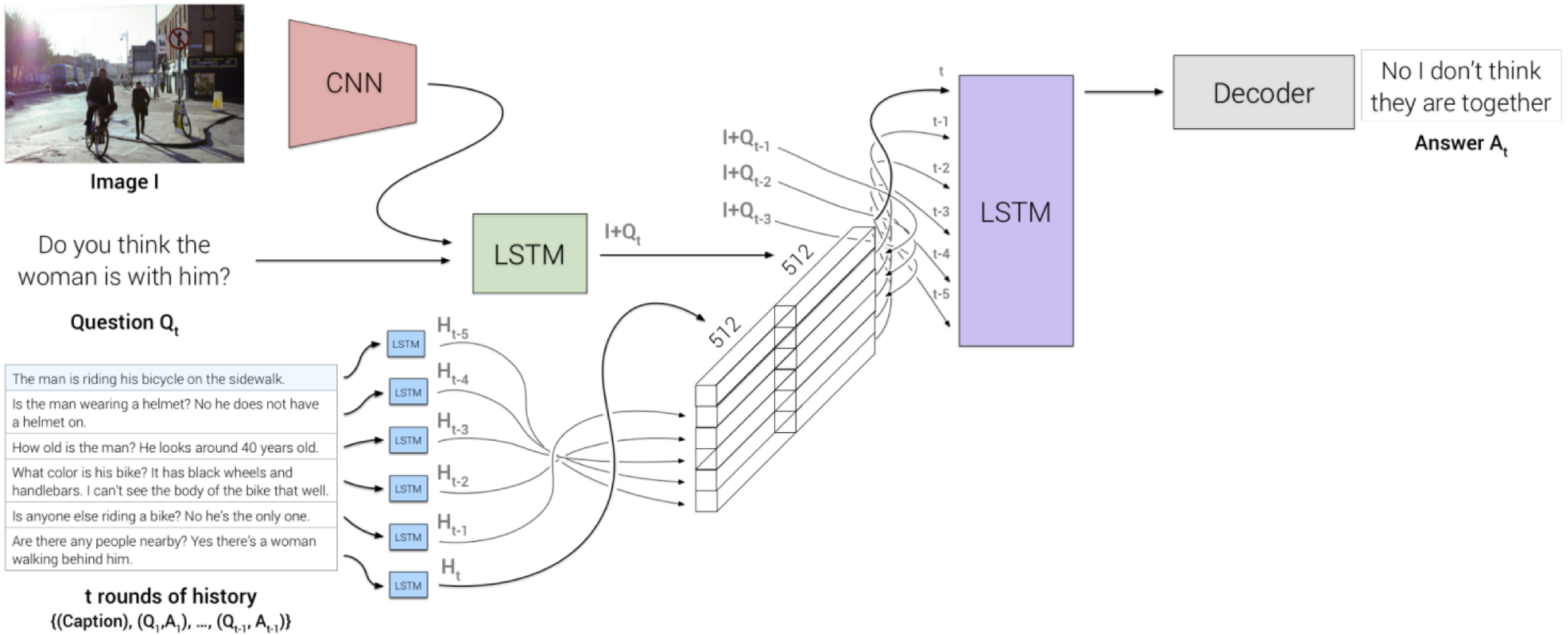


Figure 13: Hierarchical Recurrent Encoder (HRE).

8.5 Visual Dialog

- **Memory Network**: the image is encoded with a VGG-16 CNN and question with an LSTM
- the representations are concatenated and followed by a fully-connected layer and tanh nonlinearity to get a “query vector”
- each caption/QA-pair (or “fact”) in dialog history is encoded independently by an LSTM with shared weights
- the query vector is then used to compute attention over the t facts by inner product
- convex combination of attended history vectors is passed through a fully-connected layer and tanh nonlinearity, and added back to the query vector
- this combined representation is then passed through another fully-connected layer and tanh nonlinearity and then used to decode the response
- the model architecture is shown in Figure 14

8.5 Visual Dialog

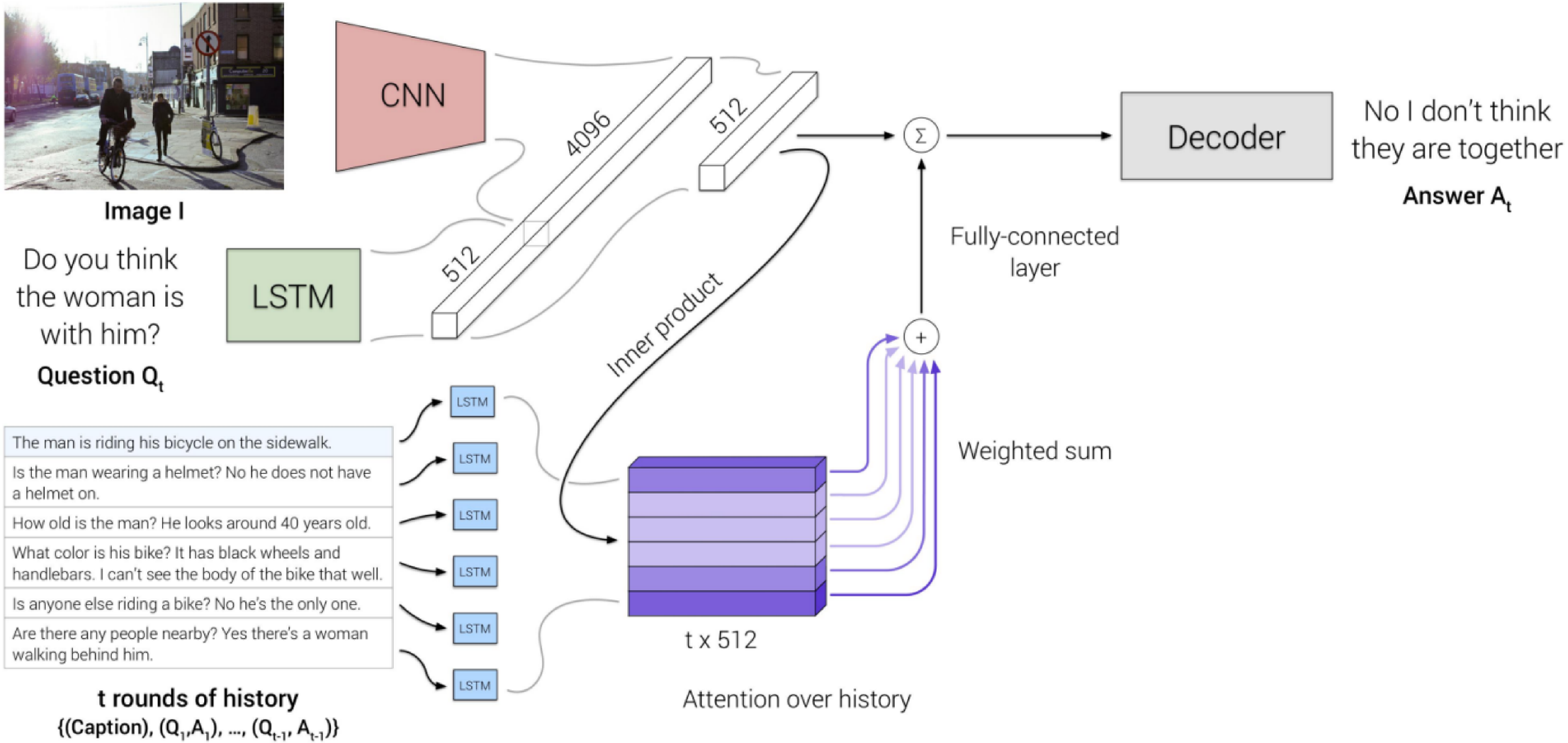


Figure 14: Memory Network Encoder (MNE).